Project Report

on

Netflix Content Analytics: An Interactive Tableau Dashboard

Subject Code: 25CAH-727

Submitted By:

Name: RIYA RAINA

UID: 25MCI10064

Class: MCA AIML

Section: MAM-2B

Submitted To:

Mr. Sanjay Kumar Aggarwal

Masters in Computer Applications (MCA)

University Institute of Computing

Chandigarh University,

Gharuan, Mohali, Punjab, India-140413

UNIVERSITY INSTITUTE of COMPUTING
Asia's Fastest Growing University

NAAC GRADE A+
Accredited University

CERTIFICATE

This is to certify that Riya Raina (25MCI10064), a student of at Chandigarh University, has completed the project titled **"Netflix Content Analytics: An Interactive Tableau Dashboard"** under my supervision.

The project has been carried out as part of the academic curriculum and is submitted in partial fulfillment of the requirements for the award of the degree. The work presented in this report is based on the public Netflix Movies and TV Shows dataset (6,234 titles) and a Tableau workbook containing worksheets and a dashboard prepared for the project.

To the best of my knowledge, the work is original and has not been submitted elsewhere for the award of any other degree or diploma.

Project Guide
Mr. Sanjay kumar aggarwal
University Institute of Computing
Chandigarh University

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Chandigarh University and the University Institute of Computing for giving me the opportunity to carry out the project titled **"Netflix Content Analytics: An Interactive Tableau Dashboard"**.

I am deeply thankful to my project guide, **Mr. Sanjay Kumar Aggarwal** , for his guidance, patience and honest feedback, especially on structuring the analysis and keeping the dashboard readable.

I would also like to thank my friends and classmates for discussions that helped me look at the data from different angles, and my family for their constant support.

Riya Raina
25MCI10064
MCA(AI&ML)
Chandigarh University

ABSTRACT

Streaming platforms decide every day what content to add, promote and invest in, and these decisions depend on understanding what a catalogue contains. This project, **Netflix Content Analytics: An Interactive Tableau Dashboard**, uses a public dataset of Netflix titles and Tableau to turn a flat file of titles into interactive charts that can be read at a glance.

The dataset (netflix_titles.csv, from Kaggle) contains 6,234 titles described by 12 columns, of which 4,265 are movies and 1,969 are TV shows. In Tableau, six worksheets are built from this single CSV source: a ratings column chart, a top-10 genre bar chart, a filled country map, a movies-versus-TV-shows bubble chart, a yearly area chart and a title detail panel. The worksheets are combined into one dashboard with a black and red theme and dashboard actions.

The figures shown by the dashboard were cross-checked. The catalogue is mainly films (68.42%). The United States (2,032 titles) and India (777) supply the most titles, and TV-MA (2,027) and TV-14 (1,698) together make up nearly 60% of the content. Documentaries (299 titles) and Stand-Up Comedy (273) are the most frequent genre categories, and additions to the library peaked in 2019 with 2,349 titles. The fall shown for 2020 is caused by incomplete data, because the file was captured a few weeks into 2020. The report documents the dataset, the design and implementation of the dashboard, the screenshots with their explanations, the checks carried out, the limitations, and possible future improvements.

Keywords: Tableau, Data Visualization, Netflix, Content Analytics, Dashboard, Business Intelligence.

UNIVERSITY INSTITUTE of COMPUTING
Asia's Fastest Growing University

NAAC GRADE A+
Accredited University

TABLE OF CONTENTS

Abstract iii

CHAPTER 1: INTRODUCTION 1

1.1 Project Overview 1
1.2 Aim of the Project 1
1.3 Objectives 1
1.4 Motivation 1
1.5 Scope of the Project 1
1.6 Importance of Data Visualization 2

CHAPTER 2: DATASET AND TOOLS 3

2.1 Introduction to the Netflix Dataset 3
2.2 Dataset Description 3
2.3 Important Fields 3
2.4 Software and Tools Used 4
2.5 Tableau Workbook and Data-Source Description 4

CHAPTER 3: SYSTEM DESIGN AND IMPLEMENTATION 6

3.1 Project Workflow 6
3.2 Connecting the Dataset to Tableau 6
3.3 Dimensions and Measures 6
3.4 Data Preparation 6
3.5 Creating the Worksheets 7
3.6 Designing and Arranging the Dashboard 7
3.7 Chart Types and Their Purposes 8
3.8 Filters, Legends and Interactive Features 8

CHAPTER 4: SCREENSHOTS, RESULTS AND DISCUSSION 9

4.1 How to Read This Chapter 9
4.2 Dashboard Overview 9
4.3 Visualizations on the Dashboard 10

CHAPTER 5: TESTING AND RESULTS 15

5.1 Data-Source Verification 15
5.2 Validation of Important Measures 15
5.3 Chart and Dashboard Checks 16
5.4 Results and Observations 17
5.5 Dataset Limitations 17
5.6 Challenges Encountered 17

CHAPTER 6: CONCLUSION AND FUTURE SCOPE 19

6.1 Summary of the Project 19
6.2 Achievements 19
6.3 Usefulness of the Dashboard 19
6.4 Future Scope 19

REFERENCES 20

LIST OF TABLES

Table 1: Summary of the dataset 3
Table 2: Important fields of the netflix_titles.csv file 4
Table 3: Software and tools 4
Table 4: Worksheets and dashboard in the Tableau workbook 5
Table 5: Project workflow 6
Table 6: Missing values and how they were handled 7
Table 7: Fields and purpose of each worksheet 7
Table 8: Chart types and what each is used for 8
Table 9: Key figures shown on the dashboard 10
Table 10: Movies and TV shows (reported from the dashboard) 10
Table 11: Ten countries with the most titles (reported from the dashboard) 11
Table 12: Titles by rating (reported from the dashboard) 12
Table 13: Titles added by year (reported from the dashboard) 14
Table 14: Data-source checks 15
Table 15: Comparison of values in the screenshots with the dataset summary 16
Table 16: Chart and dashboard checks 16
Table 17: Summary of key findings 17

CHAPTER 1: INTRODUCTION

1.1 Project Overview

This project, Netflix Content Analytics: An Interactive Tableau Dashboard, presents an analysis of the Netflix catalogue through interactive charts. The data comes from the public Netflix Movies and TV Shows file (netflix_titles.csv) published on Kaggle, which lists the titles available on the platform together with their type, country, rating, genre and date added.

A Tableau workbook was created for the project. It connects to the CSV file, contains six worksheets and combines them into one dashboard. The dashboard answers five questions quickly:

- **Type:** how much of the catalogue is film and how much is television?
- **Country:** which countries contribute the most titles?
- **Rating:** which age ratings dominate?
- **Genre:** which genres are most common?
- **Growth:** how quickly did the library expand over the years?

1.2 Aim of the Project

The aim of the project is to design a Tableau dashboard that presents the structure of the Netflix catalogue clearly, so that the main patterns in content type, country, rating, genre and growth can be understood quickly without reading the raw data.

1.3 Objectives

- To explore the dataset: study the 12 columns, their data types and missing values.
- To compare movies and TV shows: measure the split and show it in a distribution chart.
- To analyse content by country: build a world map of titles per country.
- To study ratings: count titles under each maturity rating to see the target audience.
- To identify popular genres: rank and show the ten most common genre categories.
- To track growth over time: plot titles added per year, split by movies and TV shows.
- To design a usable dashboard: combine all worksheets with a consistent theme and a title-detail panel.
- To document the project, its results and its limitations in a report.

1.4 Motivation

Streaming platforms decide every day what content to add, promote and invest in. Netflix is the best-known example, with a library spanning many countries, languages and genres. A raw CSV file with more than six thousand rows cannot be understood by scrolling, and totals alone hide the differences between countries, ratings and years.

Tableau lets an analyst connect a file, drag fields onto a canvas and see results immediately, which is faster than writing code. The Netflix dataset is a good practice ground because it is large, mixes text, number, date and geographic fields, and is untidy like real data (blanks, multi-value cells, dates stored as text).

1.5 Scope of the Project

The scope of the project covers the following.

- One data source: the Netflix Movies and TV Shows CSV file (netflix_titles.csv) with 6,234 rows and 12 columns.
- Six worksheets and one dashboard in a single Tableau workbook.
- Analysis of titles by type, country, rating, genre and year added.
- Checking of the displayed figures against each other and against the dataset summary.

The project is limited to descriptive analytics: it answers "what does the catalogue contain?". It does not include predicting viewership, recommending titles or live data connections, because the data has no views, watch time or user ratings. The file was captured in early 2020, so the results are a snapshot, not today's Netflix library. These points are discussed as future scope in Chapter 6.

1.6 Importance of Data Visualization

Data visualization turns numbers into shapes, positions and colours that the human eye can compare quickly. A list of 6,234 titles says very little, but a bar chart of ratings or a map of countries shows the pattern at once. In the entertainment business, such patterns guide content acquisition, regional library planning and audience targeting.

In a business setting, a dashboard brings several views of the same data onto one screen. A manager can compare content type, country, rating, genre and growth together instead of reading separate reports, which supports faster and better-informed decisions.

CHAPTER 2: DATASET AND TOOLS

2.1 Introduction to the Netflix Dataset

The Netflix Movies and TV Shows dataset describes the titles in the library of Netflix, a streaming service with content from many countries, languages and genres. It is a public file (netflix_titles.csv) published on Kaggle, and the version used in this project has 6,234 rows. The data is a snapshot captured in early 2020 and has no viewing figures, so the results show what the catalogue contains and not how often the titles are watched.

2.2 Dataset Description

The dataset is a single CSV file, summarised in Table 1. It has one row for each title. Show ID is unique per row, so a distinct count of Show ID gives the number of titles.

Table 1: Summary of the dataset

Property	Value
File name	netflix_titles.csv
Total records / columns	6,234 rows (one per title) / 12 columns
Movies	4,265 (68.42%)
TV shows	1,969 (31.58%)
Release years covered	1925 to 2020
Date added range	2008 to 2020
Distinct countries	111 (after splitting multi-country entries)

The titles were released between 1925 and 2020 and were added to Netflix between 2008 and 2020. Several columns (Cast, Country and Listed In) hold more than one comma-separated value in a cell, and some columns contain blanks, which are examined in Section 3.4.

2.3 Important Fields

Table 2 lists the 12 columns of the file and how each is used in the Tableau worksheets. In Tableau, text and date fields are normally treated as dimensions, and numeric fields that are counted or summed are treated as measures.

Table 2: Important fields of the netflix_titles.csv file

Field	Data type	Role	Description / use in the project
Show Id	Number	Measure	Unique key; distinct count gives the number of titles
Type	Text	Dimension	Movie or TV Show; used as colour in the bubble and area charts
Title	Text	Dimension	Shown in the title detail panel
Director, Cast	Text	Dimension	Not used in charts (many blanks)
Country	Text	Dimension	Given a geographic role; used in the country map
Date Added	Date	Dimension	Converted from text; used in the yearly chart and detail panel
Release Year	Number	Dimension	Shown in the title detail panel
Rating	Text	Dimension	Used in the ratings chart
Duration	Text	Dimension	Shown in the detail panel (minutes or seasons)
Listed In	Text	Dimension	Genre categories; used in the top 10 genre chart
Description	Text	Dimension	Shown in the title detail panel

The measure used in every worksheet is CNTD([Show Id]), the distinct count of titles. The dimensions Type, Country, Rating, Listed In and Date Added decide how this measure is grouped. Director and Cast are available in the data source but are not placed in any worksheet.

2.4 Software and Tools Used

Table 3: Software and tools

Tool	Purpose
Tableau Desktop	Connecting to the data, building the worksheets and the dashboard using drag-and-drop and built-in calculations
netflix_titles.csv	The single data source: 6,234 rows and 12 columns, read through a text-file connection
Mapbox and OpenStreetMap	Base map tiles behind the country map (visible in the credit line of the map); a dark style was used to match the theme
Microsoft Word	Preparation of this report; screenshots were exported from Tableau as images

2.5 Tableau Workbook and Data-Source Description

The workbook has one data source, the CSV file netflix_titles.csv, connected through Tableau's "Text file" option. Tableau lists the 12 fields under Dimensions and Measures. The main measure is CNTD([Show Id]), and Country was given a geographic role so that Tableau generates latitude and longitude for the map. Filters and sorting limit the genre chart to the top ten categories.

The workbook contains six worksheets and one dashboard. Table 4 lists them.

Table 4: Worksheets and dashboard in the Tableau workbook

Worksheet	Dashboard	Chart type
Ratings chart	Netflix dashboard	Column chart, sorted high to low
Top 10 genres	Netflix dashboard	Ranked horizontal bar chart, top 10 categories
Country map	Netflix dashboard	Filled map of countries
Movies vs TV shows	Netflix dashboard	Bubble chart by Type
Titles by year	Netflix dashboard	Area chart, split by Type
Title detail panel	Netflix dashboard	Text panel

CHAPTER 3: SYSTEM DESIGN AND IMPLEMENTATION

3.1 Project Workflow

The project follows the steps shown in Table 5, from obtaining the data to presenting the dashboard.

Table 5: Project workflow

Step	Activity
1	Obtain the Netflix Movies and TV Shows CSV file (netflix_titles.csv) from Kaggle
2	Connect the CSV file to Tableau using the Text file option
3	Review the 12 columns and identify dimensions and measures
4	Count the missing values in every column
5	Prepare the fields: convert Date Added to a date and give Country a geographic role
6	Build individual worksheets, one chart for each question
7	Format the worksheets with colour, labels, sorting and filters
8	Combine the worksheets into the dashboard, apply the dark theme and add dashboard actions
9	Capture screenshots of the dashboard and its charts
10	Check the displayed values against each other and against the dataset summary, and document the results

3.2 Connecting the Dataset to Tableau

In Tableau Desktop, the CSV file was connected through the "Text file" option. Tableau read the 12 columns and listed the fields under Dimensions and Measures. Some columns are text (Title, Description), some are numeric (Show Id, Release Year), one is a date (Date Added), and a few hold several comma-separated values (Cast, Country, Listed In).

3.3 Dimensions and Measures

Tableau divides fields into dimensions, which are used to split the data into groups, and measures, which are aggregated. In this project the main measure is the distinct count of Show Id, CNTD([Show Id]), and every worksheet uses it so that the numbers are directly comparable. Show Id is numeric but is never summed or averaged.

The dimensions Type, Country, Rating, Listed In and Date Added decide how the count is grouped. For example, CNTD([Show Id]) by Rating gives the number of titles under each rating, and CNTD([Show Id]) by Country gives the number of titles of each country.

3.4 Data Preparation

Blank cells were counted in every column before any chart was built, so that the charts would not be misread. Table 6 shows the missing values and how each case was handled.

Table 6: Missing values and how they were handled

Column	Missing	How it was handled
Director	1,969	Left as is; field not used in charts
Cast	570	Left as is; field not used in charts
Country	476	Titles without a country do not appear on the map
Date Added	11	Excluded automatically from the yearly chart
Rating	10	Shown as "Null" in the ratings chart (kept on purpose so that totals match the dataset)
Other columns	0	No action needed

The following points are also relevant to preparation.

- Date Added was stored as text (for example "September 9, 2019") and was converted to a date field so that it could be grouped by year.
- Country received the Country/Region geographic role, so no latitude or longitude data was added.
- Show Id was treated as a number but was only used through a distinct count.
- No calculated field other than the distinct count of titles was needed for the worksheets.

3.5 Creating the Worksheets

Each worksheet answers one question about the catalogue. Table 7 gives the fields used in each worksheet and its purpose.

Table 7: Fields and purpose of each worksheet

Worksheet	Fields used	Purpose
Ratings chart	Rating on Columns; CNTD(Show Id) on Rows; bars sorted high to low, red, with value labels	Show which age ratings dominate
Top 10 genres	Listed In on Rows; CNTD(Show Id) on Columns; top-10 filter by count, sorted descending	Rank the ten most common genre categories
Country map	Country; CNTD(Show Id) as colour, so more titles give stronger red; dark Mapbox base	Show which countries contribute the most titles
Movies vs TV shows	Type as colour; CNTD(Show Id) as size and label	Show the movie and TV show split with count and percentage
Titles by year	Date Added (year) on Columns; CNTD(Show Id) on Rows; Type as colour; Area mark	Show how the library grew each year, split by type
Title detail panel	Title, Type, Rating, Duration, Release Year, Date Added, Listed In and Description as text	Show the details of a selected title

3.6 Designing and Arranging the Dashboard

The six worksheets were placed on one dashboard of fixed size, so that the layout stays identical on any screen. The title detail panel and the Netflix logo are on top. The country map, the ratings chart and the movies-versus-TV-shows bubbles form the middle band. The top 10 genre chart and the titles-by-year chart are at the bottom.

The dashboard uses a black background with red titles and bars, red gradients on the map and bubbles, and light axis labels for legibility, so that it matches the Netflix identity.

3.7 Chart Types and Their Purposes

Table 8: Chart types and what each is used for

Chart type	Best used for	Used in this project to
Column chart	Comparing values across categories	Compare the number of titles under each rating
Horizontal bar chart	Comparing values across categories with long labels	Rank the ten most common genre categories
Filled map	Showing values by place	Show titles per country
Bubble chart	Showing parts of a whole with few categories	Show the movie and TV show split
Area chart	Showing change over time	Show titles added per year, split by type
Text panel	Showing the details of one item	Show the details of a selected title

3.8 Filters, Legends and Interactive Features

Only features supported by the project description or by the screenshots are described here.

- **Worksheet filters.** The Top 10 genres worksheet is filtered to the ten categories with the highest count and sorted in descending order. The yearly chart leaves out the 11 titles that have no Date Added.
- **Dashboard actions.** Actions filter or highlight related views and update the title detail panel when a title is clicked.
- **Tooltips.** Tooltips show the exact numbers when the pointer is placed over a chart.
- **Legends.** A Type legend (Movie, TV Show) and a colour legend for the distinct count are placed beside the bubble chart. The Movie and TV Show layers of the area chart are labelled inside the chart.
- **Data labels.** Counts are displayed above the bars of the ratings chart, and the count and percentage are displayed inside each bubble.
- **Shortened labels.** Because the panels are small, the rating names on the horizontal axis and some long genre names are shortened with ".." in the screenshot.

CHAPTER 4: SCREENSHOTS, RESULTS AND DISCUSSION

4.1 How to Read This Chapter

This chapter presents the screenshots from the project and explains each visualization. To keep what was seen separate from what was reported, two labels are used.

- **Observed in the screenshot:** what is visible in the supplied screenshot, such as labels, shapes, legends and relative sizes.
- **Reported from the dashboard:** the counts displayed by the dashboard, together with simple sums and percentages calculated from them.

Where a screenshot value was compared with another value, the result is given in Chapter 5. All screenshots are the original images exported from Tableau and are not edited.

4.2 Dashboard Overview

Figure 1 shows the complete dashboard, which contains six worksheets.

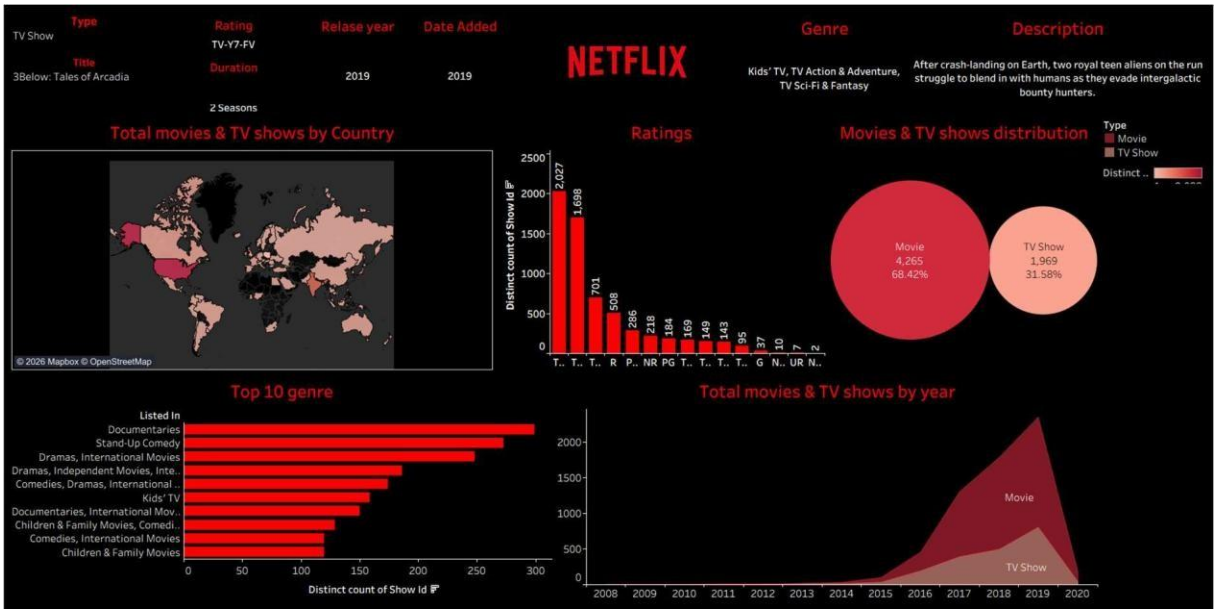


Figure 1: Complete Netflix dashboard

Observed in the screenshot: The dashboard has a black background and red titles. The top strip holds the title detail panel, with the headings Type, Title, Rating, Duration, Release Year, Date Added, Genre and Description, and the NETFLIX logo in the centre. The middle band holds the world map "Total movies & TV shows by Country", the "Ratings" column chart and the "Movies & TV shows distribution" bubble chart with its legends at the top right. The lower band holds the "Top 10 genre" bar chart and the "Total movies & TV shows by year" area chart. The individual panels are explained in Section 4.3.

Purpose: To show the structure of the whole catalogue on one screen, so that type, country, rating, genre and growth can be compared together.

Reported from the dashboard: The main figures on the dashboard are given in Table 9.

Table 9: Key figures shown on the dashboard

Measure	Value
Titles in the catalogue	6,234
Movies (share of titles)	68.42%
United States titles	2,032
TV-MA titles	2,027
Peak year of additions	2019 (2,349 titles)

4.3 Visualizations on the Dashboard

4.3.1 Movies vs TV Shows (Bubble Chart)

Observed in the screenshot: Two circles on a black background. A large dark-red circle for Movie is labelled 4,265 and 68.42%, and a smaller light-red circle for TV Show is labelled 1,969 and 31.58%. A Type legend (Movie, TV Show) and a colour legend for the distinct count are at the top right, and the label of the colour legend is cut off.

Purpose: To show how much of the catalogue is film and how much is television.

Reported from the dashboard: The catalogue is mainly films: 4,265 movies (68.42%) against 1,969 TV shows (31.58%), so a little over two in three titles are movies (Table 10). TV shows have many episodes, so this count understates their viewing time.

4.3.2 Titles by Country (Map)

Figure 2 shows the country map.



Figure 2: Total movies and TV shows by country

Observed in the screenshot: A world map on a dark Mapbox base, with countries in shades of red. The United States is the darkest, India is shaded orange-red, and most other countries are light pink. Parts of Africa and Central Asia appear almost black. The credit line "© 2026 Mapbox © OpenStreetMap" is at the bottom left.

Purpose: To show which countries contribute the most titles.

Reported from the dashboard: The United States leads with 2,032 titles, about a third of the catalogue, followed by India (777), the United Kingdom (348), Japan (176) and Canada (141). South Korea (136), Spain (117), France (90), Mexico (83) and Turkey (79) complete the top ten (Table 11). Multi-country titles count under each country, and the 476 titles with no country are not on the map.

Table 11: Ten countries with the most titles (reported from the dashboard)

Rank	Country	Titles
1	United States	2,032
2	India	777
3	United Kingdom	348
4	Japan	176
5	Canada	141
6	South Korea	136
7	Spain	117
8	France	90
9	Mexico	83
10	Turkey	79

4.3.3 Ratings (Column Chart)

Observed in the screenshot: A column chart in red with the distinct count of Show Id on the vertical axis (0 to about 2,500) and the rating on the horizontal axis, sorted from highest to lowest. Every bar has a value label (2,027, 1,698, 701, 508, 286, 218, 184, 169, 149, 143, 95, 37, 10, 7 and 2). Some rating names on the horizontal axis are shortened because the panel is narrow.

Purpose: To show which age ratings dominate the catalogue and therefore which audience it targets.

Reported from the dashboard: TV-MA (2,027) and TV-14 (1,698) together make 3,725 titles, which is 59.8% of the catalogue, so Netflix content mainly targets teenagers and adults. Next are TV-PG (701), R (508), PG-13 (286), NR (218) and PG (184). Child-oriented ratings are much smaller: TV-Y7 (169), TV-G (149), TV-Y (143) and TV-Y7-FV (95). The rarest are G (37), UR (7) and NC-17 (2), and 10 titles have no rating. Table 12 gives all values.

Table 12: Titles by rating (reported from the dashboard)

Rating	Titles	Share of titles
TV-MA	2,027	32.5%
TV-14	1,698	27.2%
TV-PG	701	11.2%
R	508	8.1%
PG-13	286	4.6%
NR	218	3.5%
PG	184	3.0%
TV-Y7	169	2.7%
TV-G	149	2.4%
TV-Y	143	2.3%
TV-Y7-FV	95	1.5%
G	37	0.6%
No rating (Null)	10	0.2%
UR	7	0.1%
NC-17	2	0.0%
Total	**6,234**	**100.0%**

4.3.4 Top 10 Genres (Bar Chart)

Figure 3 shows the top 10 genre categories.

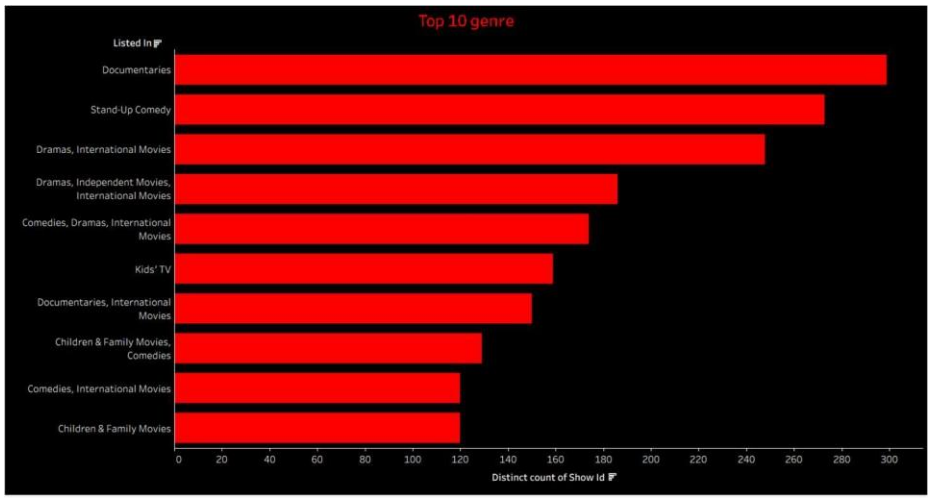


Figure 3: Top 10 genre categories

Observed in the screenshot: Ten horizontal red bars sorted from the longest to the shortest, with the genre categories on the vertical axis and the distinct count of Show Id (0 to about 300) on the horizontal axis. Some long category names are shortened with "..".

Purpose: To show the ten most common genre categories.

Reported from the dashboard: Documentaries lead with 299 titles, closely followed by Stand-Up Comedy (273) and Dramas with International Movies (248). The remaining categories are mixed combinations with 186 down to 120 titles each, such as Kids' TV (159) and Documentaries with International Movies (150).

Note: the Listed In column stores several genres in one cell, and the chart counts each unique combination as one category. The bars therefore show how often a combination appears, not a single genre. International Movies appearing in many combinations still points to the global reach of Netflix.

4.3.5 Titles Added by Year (Area Chart)

Figure 4 shows the number of titles added each year.

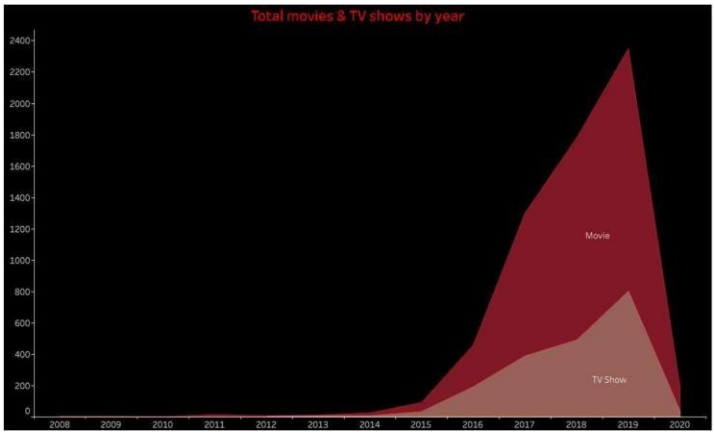


Figure 4: Total movies and TV shows added by year

Observed in the screenshot: An area chart with the years 2008 to 2020 on the horizontal axis and the count (up to about 2,000 and above) on the vertical axis. Two layers are drawn: dark red for Movie and a lighter shade for TV Show, each labelled inside the chart. The areas stay close to zero until about 2015, rise steeply to a peak in 2019 and fall sharply in 2020.

Purpose: To show how quickly the library expanded over the years and how the movie and TV show shares changed.

Reported from the dashboard: Activity is very low before 2015, then rises fast: about 456 titles in 2016, 1,300 in 2017, 1,780 in 2018 and a peak of 2,349 in 2019 (1,546 movies and 803 TV shows). The fall in 2020 (147 movies and 37 TV shows) is incomplete data and not a slowdown, because the file was captured only a few weeks into 2020 (Table 13).

Table 13: Titles added by year (reported from the dashboard)

Year added	Titles	Remark
Before 2015	Very low	Few titles were added each year
2016	About 456	Start of rapid growth
2017	About 1,300	Growth continues
2018	About 1,780	Growth continues
2019	2,349	Peak year: 1,546 movies and 803 TV shows
2020	184	Incomplete year: 147 movies and 37 TV shows

4.3.6 Title Detail Panel (Text)

Observed in the screenshot: A strip along the top of the dashboard with red headings and white values for the selected title. In the screenshot the values are: Type = TV Show, Title = 3Below: Tales of Arcadia, Rating = TV-Y7-FV, Duration = 2 Seasons, Release Year = 2019, Date Added = 2019, and Genre = Kids' TV, TV Action & Adventure, TV Sci-Fi & Fantasy, with a short description of the show. The heading "Release Year" is spelled "Relase year" on the dashboard.

Purpose: To show the full details of any title that is selected in the other charts.

Reported from the dashboard: The panel combines the text fields Title, Type, Rating, Duration, Release Year, Date Added, Listed In and Description, and it updates when a title is selected through the dashboard actions.

CHAPTER 5: TESTING AND RESULTS

The project materials do not include a formal test log, so the testing in this chapter consists of checks made while preparing the report: comparing the values in the screenshots with the dataset summary and with each other, and checking the dashboard for completeness and legibility.

5.1 Data-Source Verification

Table 14: Data-source checks

Check	Expected	Result	Status
Number of columns	12	12 columns described (Director and Cast are listed together in Table 2)	Matched
Number of titles	6,234	4,265 movies + 1,969 TV shows = 6,234	Matched
Movie share	68.42%	4,265 ÷ 6,234 = 68.42%	Matched
TV show share	31.58%	1,969 ÷ 6,234 = 31.58%	Matched
Ratings add up to the total	6,234	The 15 bars of the ratings chart add up to 6,234	Matched
Titles without a rating	10	The "Null" bar of the ratings chart is 10	Matched
Date range of additions	2008 to 2020	The yearly chart axis runs from 2008 to 2020	Matched
Missing values in other columns	Recorded	Director 1,969; Cast 570; Country 476; Date Added 11	Recorded

5.2 Validation of Important Measures

The values that are visible in the screenshots were compared with the dataset summary (Table 1) and with each other. Table 15 shows the comparison.

Table 15: Comparison of values in the screenshots with the dataset summary

Measure	Value in screenshot	Value used for comparison	Status
Movies (bubble chart)	4,265	4,265 (Table 1)	Matched
TV shows (bubble chart)	1,969	1,969 (Table 1)	Matched
Movie percentage	68.42%	68.42% (4,265 ÷ 6,234)	Matched
TV show percentage	31.58%	31.58% (1,969 ÷ 6,234)	Matched
Highest rating (TV-MA)	2,027	2,027	Matched
Second rating (TV-14)	1,698	1,698	Matched
TV-MA + TV-14	2,027 + 1,698	3,725 (59.8% of 6,234)	Matched
Titles with no rating	10	10 (Table 6)	Matched
Top genre (Documentaries)	Bar just below 300	299	Consistent
Peak year 2019	Area peaks in 2019	1,546 + 803 = 2,349	Matched
Year 2020	Sharp fall	147 + 37 = 184 (partial year)	Explained
Top country	United States darkest on map	2,032 (highest count)	Consistent
Second country	India shaded next	777 (second highest count)	Consistent

All values compared agree with each other and with the dataset summary. These checks compare the dashboard with the figures reported for the dataset; the same figures can be re-checked at any time by counting the distinct Show Id values in netflix_titles.csv.

5.3 Chart and Dashboard Checks

Table 16: Chart and dashboard checks

Item checked	Observation
Dashboard contains all its worksheets	Passed: six panels are visible (detail panel, map, ratings, distribution, genres and yearly chart)
Chart type matches the purpose	Passed: column and bar charts for comparison, bubbles for the two-part split, area chart for change over time, map for geography
Colours and legends	Passed: the black and red theme is consistent in all charts. A Type legend is present. The colour legend beside the bubbles is partly cut off
Titles on panels	Passed: all five charts are titled
Label legibility	Partly met: rating names on the horizontal axis and some genre names are shortened with ".."
Map legend visible	Not met: the colour legend for the map is not visible in the screenshot
Explanation of 2020 drop	Not met: the yearly chart has no note saying that 2020 is an incomplete year
Spelling of headings	Not met: "Release year" is spelled "Relase year" in the detail panel

5.4 Results and Observations

The main findings of the dashboard are summarised in Table 17.

Table 17: Summary of key findings

Area	Finding
Content type	68.42% movies, 31.58% TV shows
Top country	United States, 2,032 titles
Second country	India, 777 titles
Most common rating	TV-MA, 2,027 titles
TV-MA + TV-14	3,725 titles, nearly 60% of the catalogue
Top genre category	Documentaries, 299 titles
Peak year of additions	2019, with 2,349 titles
Average movie length	About 99 minutes (from the Duration column)
TV shows with one season	About 67% of all TV shows (from the Duration column)

The main observations are as follows.

- **Content type.** The library is dominated by movies (68.42%).
- **Countries.** The United States and India together supply 2,809 titles, about 45% of the catalogue. Large parts of Africa and Central Asia have few or no titles.
- **Ratings.** TV-MA and TV-14 make up almost 60% of the content, while child-oriented ratings are much smaller.
- **Genres.** Documentaries and Stand-Up Comedy are the most frequent genre categories.
- **Growth.** The catalogue expanded rapidly after 2015 and peaked in 2019; the 2020 drop reflects when the data was captured.

5.5 Dataset Limitations

- The dataset is a snapshot from early 2020, so later additions are not included.
- Genres are stored as combinations, so single genres cannot be counted without splitting the field.
- There is no viewership, popularity or user-rating data, so the performance of titles cannot be measured.
- Titles with a missing country (476) or rating (10) are not fully represented, and Director (1,969 blanks) and Cast (570 blanks) are too incomplete to analyse.
- Titles with several countries are counted under each country, so the country counts add up to more than the number of titles.

5.6 Challenges Encountered

No record of problems met during development was supplied, so only issues that are evident from the project materials are noted.

- Date Added was stored as text and had to be converted to a date, and blank values in several columns had to be counted and handled so that the charts would not be misread.
- Cast, Country and Listed In hold several values in one cell, so counts for single countries and genres are not straightforward.
- Fitting long category names and many ratings into small panels led to shortened labels.
- The yearly chart shows a sharp fall in 2020 that could be misread as a slowdown if the capture date of the data is not known.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Summary of the Project

This project used Tableau to visualize and analyse the Netflix Movies and TV Shows dataset. The CSV file was connected as the data source, six worksheets were built from it, and these were arranged into one dashboard. The dashboard presents the catalogue by type, country, rating, genre and year added. The values shown in the dashboard were checked against each other and against the dataset summary of 6,234 titles.

The main findings are that the library is dominated by movies (68.42%), that the United States and India together supply about 45% of all titles, that TV-MA and TV-14 make up almost 60% of the content, that Documentaries and Stand-Up Comedy are the most frequent genre categories, and that the catalogue expanded rapidly after 2015 and peaked in 2019.

6.2 Achievements

- A Tableau workbook with one data source, six worksheets and one dashboard was completed.
- A map, column, bar, bubble and area chart and a text panel were used, each matched to a specific analytical purpose.
- The values visible in the screenshots were checked and agree with the dataset summary.
- The analysis identified the content mix (movies and TV shows), the leading countries, the dominant ratings, the common genre categories and the peak year of additions.
- The work gave hands-on experience of the whole analytics workflow: understand, clean, visualise and communicate.

6.3 Usefulness of the Dashboard

The dashboard allows a viewer to answer common questions about the catalogue at a glance: how much of it is film, which countries contribute the most, which audiences it targets, which genres are common and when the library grew. Placing related charts together makes comparison easy, and the consistent colour theme keeps the views uniform. Small design choices such as sorting bars, colour scales and labels make a clear difference to how easily a chart is understood. Such a dashboard can help content teams decide what to acquire, where to plan regional libraries and which audiences to target.

6.4 Future Scope

- Split the multi-value genre, country and cast fields to count each value separately.
- Add newer data to see how the catalogue changed after 2020.
- Apply text analysis to the descriptions to group titles by theme, and train a model to suggest similar titles.
- Add a visible legend to the map, a note that 2020 is an incomplete year, and correct the spelling of "Release year" in the detail panel.
- Add global filters (for example Type, Rating and Year) to the dashboard.
- Publish the dashboard on Tableau Public as a shareable interactive link.

REFERENCES

[1] Kaggle. Netflix Movies and TV Shows (netflix_titles.csv). Available at: https://www.kaggle.com/datasets/shivamb/netflix-shows

[2] Tableau Software. Tableau Help (Desktop, Dashboards, Maps, Aggregate Functions). Available at: https://help.tableau.com/

[3] Tableau Public. Available at: https://public.tableau.com/

[4] Mapbox. Available at: https://www.mapbox.com/

[5] OpenStreetMap. Copyright and License. Available at: https://www.openstreetmap.org/copyright

[6] TV Parental Guidelines. Available at: http://www.tvguidelines.org/

[7] Motion Picture Association. Film Ratings. Available at: https://www.motionpicture.org/film-ratings/

[8] Tableau workbook prepared for this project, containing the worksheets and the dashboard analysed in this report.